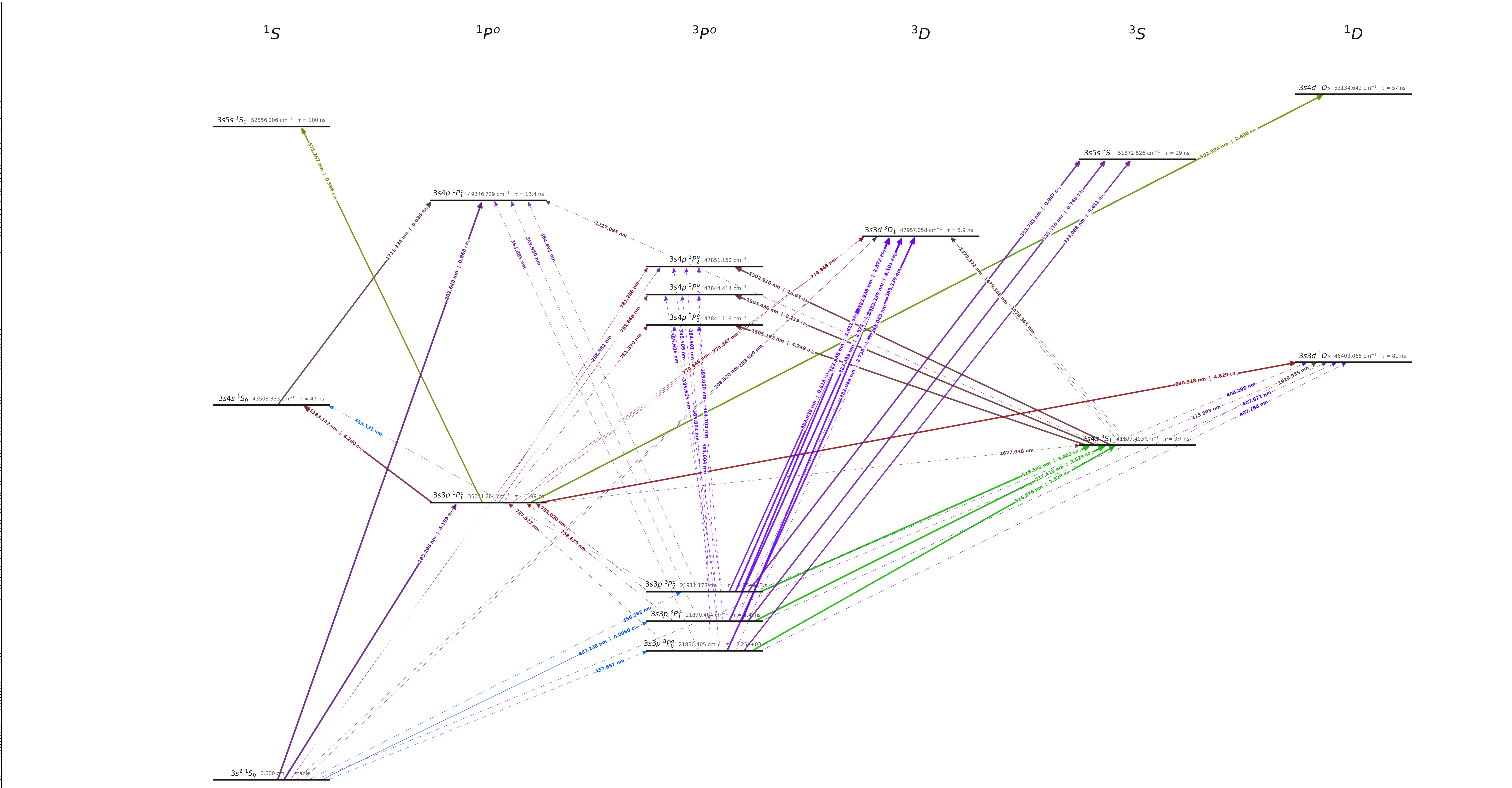


# Magnesium-24

<sup>24</sup>Mg (Mg I), nuclear spin *I* = 0. Energy levels and transitions below 2 μm: 18 levels, 61 lines.

Mg <sup>+</sup> ionisation limit 61671.05 cm<sup>-1</sup> = 7.64624 eV (162.151 nm)



## Key transitions

| Transition                                             | λ vac (nm) | λ air (nm) | ν (THz)        | U  er  U'  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                          |
|--------------------------------------------------------|------------|------------|----------------|-------------------------------|----------------------|----------------|------------|--------------------------------------------------------------|
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s3p 1P <sub>1</sub> | 285.2963   | 285.2125   | 1050.810732(5) | 4.109                         | 4.910e+08            | 79.98          | 97.7       | singlet cooling / MOT / imaging line                         |
| 3s3p 3P <sub>0</sub> - 3s3d 3D <sub>1</sub>            | 383.0441   | 382.9355   | 782.6578       | 2.735                         | 8.990e+07            | 26.98          | 53         | triplet MOT / repumping                                      |
| 3s3p 3P <sub>0</sub> - 3s3d 3D <sub>2</sub>            | 383.0446   | 382.9359   | 782.6568       | -                             | 2.110e-05            | 26.98          | 0          | triplet MOT / repumping                                      |
| 3s3p 3P <sub>1</sub> - 3s3d 3D <sub>1</sub>            | 383.3387   | 383.2299   | 782.0564       | 2.371                         | 6.740e+07            | 26.98          | 39.8       | triplet MOT / repumping                                      |
| 3s3p 3P <sub>1</sub> - 3s3d 3D <sub>3</sub>            | 383.3389   | 383.2301   | 782.0560       | -                             | 2.200e-05            | 26.98          | 0          | triplet MOT / repumping                                      |
| 3s3p 3P <sub>1</sub> - 3s3d 3D <sub>2</sub>            | 383.3391   | 383.2304   | 782.0555       | 4.101                         | 1.210e+08            | 26.98          | 71.4       | triplet MOT / repumping                                      |
| 3s3p 3P <sub>2</sub> - 3s3d 3D <sub>1</sub>            | 383.9379   | 383.8290   | 780.8358       | 0.613                         | 4.480e+06            | 26.98          | 2.64       | triplet MOT / repumping                                      |
| 3s3p 3P <sub>2</sub> - 3s3d 3D <sub>3</sub>            | 383.9381   | 383.8292   | 780.8355       | 5.611                         | 1.610e+08            | 26.98          | 95         | triplet MOT / repumping                                      |
| 3s3p 3P <sub>2</sub> - 3s3d 3D <sub>2</sub>            | 383.9383   | 383.8295   | 780.8349       | 2.372                         | 4.030e+07            | 26.98          | 23.8       | triplet MOT / repumping                                      |
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s3p 3P <sub>0</sub> | 457.2377   | 457.1096   | 655.659924     | 0.006                         | 2.540e+02            | 3.617e-05      | 100        | intercombination line: narrow-line spectroscopy / Ramsey...  |
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s3p 3P <sub>0</sub> | 457.6574   | 457.5292   | 655.058647     | -                             | 5.800e-04            | 7.074e-11      | 100        | optical lattice clock transition                             |
| 3s3p 3P <sub>0</sub> - 3s4s 3S <sub>1</sub>            | 516.8761   | 516.7321   | 580.0084       | 1.520                         | 1.130e+07            | 16.41          | 11.2       | triplet optical pumping / repumping, metastable-beam spec... |
| 3s3p 3P <sub>1</sub> - 3s4s 3S <sub>1</sub>            | 517.4125   | 517.2684   | 579.4071       | 2.629                         | 3.370e+07            | 16.41          | 33.9       | triplet optical pumping / repumping, metastable-beam spec... |
| 3s3p 3P <sub>2</sub> - 3s4s 3S <sub>1</sub>            | 518.5048   | 518.3604   | 578.1865       | 3.403                         | 5.610e+07            | 16.41          | 54.9       | triplet optical pumping / repumping, metastable-beam spec... |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## Isotope shifts

| Transition                                             | λ vac (nm) | Isotopes | Shift (MHz)  |
|--------------------------------------------------------|------------|----------|--------------|
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s4p 1P <sub>1</sub> | 202.648    | 26-24    | 2918.9(16)   |
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s4p 1P <sub>0</sub> | 202.648    | 25-24    | 1526.4(16)   |
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s3p 1P <sub>1</sub> | 285.296    | 26-24    | 1414(8)      |
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s3p 1P <sub>1</sub> | 285.296    | 25-24    | 741(9)       |
| 3s3p 3P <sub>0</sub> - 3s3d 3D <sub>1</sub>            | 383.044    | 26-24    | 60(15)       |
| 3s3p 3P <sub>0</sub> - 3s3d 3D <sub>2</sub>            | 383.045    | 26-24    | 60(15)       |
| 3s3p 3P <sub>1</sub> - 3s3d 3D <sub>1</sub>            | 383.339    | 26-24    | 61(3)        |
| 3s3p 3P <sub>1</sub> - 3s3d 3D <sub>3</sub>            | 383.339    | 26-24    | 61(3)        |
| 3s3p 3P <sub>1</sub> - 3s3d 3D <sub>2</sub>            | 383.339    | 26-24    | 61(3)        |
| 3s3p 3P <sub>2</sub> - 3s3d 3D <sub>1</sub>            | 383.938    | 26-24    | 58(4)        |
| 3s3p 3P <sub>2</sub> - 3s3d 3D <sub>3</sub>            | 383.938    | 26-24    | 58(4)        |
| 3s3p 3P <sub>2</sub> - 3s3d 3D <sub>2</sub>            | 383.938    | 26-24    | 58(4)        |
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s3p 3P <sub>0</sub> | 457.238    | 26-24    | 2683.180(20) |
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s3p 3P <sub>1</sub> | 457.238    | 25-24    | 1405.20(10)  |
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s3p 3P <sub>0</sub> | 457.657    | 26-24    | 2681.467(20) |
| 3s <sup>2</sup> 1S <sub>0</sub> - 3s3p 3P <sub>0</sub> | 457.657    | 25-24    | 1403.90(10)  |
| 3s3p 3P <sub>0</sub> - 3s4s 3S <sub>1</sub>            | 516.876    | 26-24    | -391.3(17)   |
| 3s3p 3P <sub>0</sub> - 3s4s 3S <sub>1</sub>            | 516.876    | 25-24    | -205.7(15)   |
| 3s3p 3P <sub>1</sub> - 3s4s 3S <sub>1</sub>            | 517.413    | 26-24    | -390.1(14)   |
| 3s3p 3P <sub>1</sub> - 3s4s 3S <sub>1</sub>            | 517.413    | 25-24    | -209.1(13)   |
| 3s3p 3P <sub>2</sub> - 3s4s 3S <sub>1</sub>            | 518.505    | 26-24    | -394.4(8)    |
| 3s3p 3P <sub>2</sub> - 3s4s 3S <sub>1</sub>            | 518.505    | 25-24    | -205.7(8)    |

Shift = ν(first isotope) - ν(second isotope).